03 C CPP 200Q

Topic 3 C and C++ Programming - 200 MCQs

Syllabus: C tokens/identifiers/datatypes, control structures, arrays/struct/union/string/pointers, OOP concepts (class/object/inheritance/polymorphism/overloading), C++ tokens/variables/constants/operators/control/functions/parameter passing/class/constructors/destructors/overloading/inheritance/templates/exception handling. Includes output-prediction. Distribution: ~90 C, ~30 OOP theory, ~80 C++.

Total Questions: 200 | Level: PG + UGC-NET + Faculty Screening | Mix: Theory, Numerical, Application

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Q1. What is the output of: int a=5; printf("%d", a++ + ++a); assuming evaluation left-to-right with gcc giving 12? In strict ANSI C the behaviour is?

A. 12 guar

Q2. What is output of: int x=10; printf("%d %d", x++, ++x); ?

A. 10 11 g

Q3. Output of: for(i=0;i<5;i++); printf("%d",i); with i declared int i; ?

A. 0

Q4. Output of: int a[]={1,2,3,4}; printf("%d", *(a+2)); ?

A. 2

Q5. Output of: char s[]="hello"; printf("%lu", (unsigned long)strlen(s)); ?

A. 6

Q6. Output of: int x=5; if(x=0) printf("hi"); else printf("bye"); ?

A. hi bye

Q7. Output of: printf("%d", 5/2); in C (both operands int)?

A. 2

Q8. Output of: printf("%f", 5/2); ?

A. 0.0000

Q9. Output of: int n=0; while(n++<3) printf("%d",n); ?

A. 012

Q10. Output of: char c='A'; printf("%d", c+1); ?

A. B

Q11. sizeof(int) on typical 32/64-bit gcc/Linux is?

A. 1 byte

Q12. sizeof(char) in C is by definition?

A. 2 bytes

Q13. Which storage class retains value between function calls?

A. auto

Q14. Default storage class of local variables is?

A. register

Q15. Which operator has highest precedence?

A. * over +

Q16. Output of: int a=10,b=20; printf("%d", a^b); (bitwise XOR)?

A. 0

Q17. Output of: printf("%d", ~5); on 32-bit 2's complement?

A. -5

Q18. Output of: int p=5; printf("%d", p<<2); ?

A. 10

Q19. Output of: union {int i; char c[4];} u; u.i=1; printf("%d", u.c[0]); on little-endian?

A. 0

Q20. Dereferencing a NULL pointer causes?

A. Always

Q21. C tokens include?

A. Only fu

Q22. Which is NOT a valid C identifier?

A. _abc

Q23. Which is a valid constant in C?

A. Only 12

Q24. Difference between struct and union?

A. No diffe

Q25. Arrays in C are?

A. Indexed

Q26. Pointer arithmetic p+1 advances by?

A. Always

Q27. String in C is terminated by?

A. Null cha

Q28. Which function copies strings?

A. strlen

Q29. Recursion requires?

A. Global

Q30. Passing array to function in C passes?

A. By valu

Q31. Output of: int a=4,b=5; printf("%d", a*b + a/b); ?

A. 20

Q32. Output of: int i,sum=0; for(i=1;i<=7;i++) sum+=i; printf("%d",sum); ?

A. 27

Q33. Output of: printf("%d", 8 & 8); (bitwise AND)?

A. 0

Q34. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while

Q35. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 20

Q36. Which header declares malloc/free?

A. string.h

Q37. Output of: int a=4,b=7; printf("%d", a*b + a/b); ?

A. 28

Q38. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10

Q39. Output of: printf("%d", 1 & 10); (bitwise AND)?

A. 11

Q40. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while

Q41. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 8

Q42. Which header declares malloc/free?

A. math.h

Q43. Output of: int a=6,b=6; printf("%d", a*b + a/b); ?

A. 12

Q44. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 9

Q45. Output of: printf("%d", 5 & 9); (bitwise AND)?

A. 1

Q46. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q47. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 4

Q48. Which header declares malloc/free?

A. string.h

Q49. Output of: int a=8,b=3; printf("%d", a*b + a/b); ?

A. 27

Q50. Output of: int i,sum=0; for(i=1;i<=6;i++) sum+=i; printf("%d",sum); ?

A. 21

Q51. Output of: printf("%d", 9 & 9); (bitwise AND)?

A. 0

Q52. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. none

Q53. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 4

Q54. Which header declares malloc/free?

A. math.h

Q55. Output of: int a=2,b=3; printf("%d", a*b + a/b); ?

A. 7

Q56. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 6

Q57. Output of: printf("%d", 6 & 1); (bitwise AND)?

A. 7

Q58. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q59. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 9

Q60. Which header declares malloc/free?

A. stdlib.h

Q61. Output of: int a=7,b=3; printf("%d", a*b + a/b); ?

A. 24

Q62. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 25

Q63. Output of: printf("%d", 10 & 4); (bitwise AND)?

A. 14

Q64. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q65. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 5

Q66. Which header declares malloc/free?

A. stdlib.h

Q67. Output of: int a=4,b=9; printf("%d", a*b + a/b); ?

A. 36

Q68. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10

Q69. Output of: printf("%d", 6 & 8); (bitwise AND)?

A. 14

Q70. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q71. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 8

Q72. Which header declares malloc/free?

A. stdlib.h

Q73. Output of: int a=2,b=9; printf("%d", a*b + a/b); ?

A. 11

Q74. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9

Q75. Output of: printf("%d", 7 & 3); (bitwise AND)?

A. 7

Q76. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q77. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 7

Q78. Which header declares malloc/free?

A. stdlib.h

Q79. Output of: int a=3,b=6; printf("%d", a*b + a/b); ?

A. 18

Q80. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 14

Q81. Output of: printf("%d", 7 & 1); (bitwise AND)?

A. 1

Q82. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for

Q83. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 6

Q84. Which header declares malloc/free?

A. math.h

Q85. Output of: int a=9,b=2; printf("%d", a*b + a/b); ?

A. 23

Q86. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9

Q87. Output of: printf("%d", 5 & 5); (bitwise AND)?

A. 5

Q88. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-whil

Q89. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 3

Q90. Which header declares malloc/free?

A. math.h

Q91. Encapsulation primarily achieves?

A. Data hi

Q92. Inheritance represents?

A. IS-A rel

Q93. Polymorphism in OOP means?

A. Only da

Q94. Abstraction focuses on?

A. Hiding i

Q95. A class vs object?

A. Class is

Q96. Constructor is called?

A. At obje

Q97. Destructor in C++ is denoted by?

A. Hash (#

Q98. Virtual function enables?

A. Static b

Q99. Pure virtual function makes class?

A. Concre

Q100. Function overloading requires?

A. Differer

Q101. Operator overloading cannot overload?

A. ::, ., .*,

Q102. Friend function in C++?

A. Cannot

Q103. Multiple inheritance ambiguity is resolved via?

A. Macros

Q104. HAS-A relationship is implemented via?

A. Inherita

Q105. Overriding vs overloading?

A. Overloa

Q106. Which OOP pillar binds code and data together?

A. Abstrac

Q107. Which OOP pillar enables code reuse via hierarchy?

A. Abstrac

Q108. Dynamic binding is associated with?

A. Macros

Q109. An abstract class in C++ has?

A. Only st

Q110. Which OOP pillar binds code and data together?

A. Inherita

Q111. Which OOP pillar enables code reuse via hierarchy?

A. Abstrac

Q112. Dynamic binding is associated with?

A. Virtual

Q113. An abstract class in C++ has?

A. At least

Q114. Which OOP pillar binds code and data together?

A. Abstrac

Q115. Which OOP pillar enables code reuse via hierarchy?

A. Encaps

Q116. Dynamic binding is associated with?

A. Inline fu

Q117. An abstract class in C++ has?

A. No mer

Q118. Which OOP pillar binds code and data together?

A. Inherita

Q119. Which OOP pillar enables code reuse via hierarchy?

A. Inherita

Q120. Dynamic binding is associated with?

A. Virtual

Q121. Which is correct C++ input/output?

A. cin >> x

Q122. Scope resolution operator is?

A. ->

Q123. Which header for C++ strings?

A. <string.

Q124. new/delete in C++ differ from malloc/free because?

A. They ar

Q125. Default access in C++ class is?

A. public

Q126. Default access in C++ struct is?

A. none

Q127. Copy constructor is called when?

A. Every a

Q128. Which constructor has no parameters?

A. Default

Q129. Function template allows?

A. No reus

Q130. Class template parameter is declared with?

A. overloa

Q131. Exception handling keywords are?

A. do, whi

Q132. An uncaught exception causes?

A. termina

Q133. STL vector provides?

A. Associa

Q134. Which STL container is LIFO?

A. set

Q135. Inline function suggests?

A. No retu

Q136. 'this' pointer refers to?

A. Current

Q137. Pass by reference in C++ uses?

A. * only

Q138. Function overloading resolution is?

A. Link-tim

Q139. Virtual destructor is needed when?

A. Only fo

Q140. Namespace in C++ is used to?

A. Avoid n

Q141. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. error

Q142. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210

Q143. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q144. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referen

Q145. Which cast is safest for polymorphism in C++?

A. const_c

Q146. Template specialization allows?

A. Custom

Q147. Output of C++: int x=19; cout << (x>10 ? "big" : "small"); ?

A. error

Q148. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120

Q149. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q150. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL

Q151. Which cast is safest for polymorphism in C++?

A. dynami

Q152. Template specialization allows?

A. Custom

Q153. Output of C++: int x=7; cout << (x>10 ? "big" : "small"); ?

A. 0

Q154. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 2

Q155. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q156. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referer

Q157. Which cast is safest for polymorphism in C++?

A. const_c

Q158. Template specialization allows?

A. Faster

Q159. Output of C++: int x=13; cout << (x>10 ? "big" : "small"); ?

A. big

Q160. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 012345

Q161. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q162. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q163. Which cast is safest for polymorphism in C++?

A. C-style

Q164. Template specialization allows?

A. No gen

Q165. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. error

Q166. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 3

Q167. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q168. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL

Q169. Which cast is safest for polymorphism in C++?

A. dynami

Q170. Template specialization allows?

A. Only m

Q171. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. 0

Q172. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 01

Q173. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q174. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referen

Q175. Which cast is safest for polymorphism in C++?

A. const_c

Q176. Template specialization allows?

A. No gen

Q177. Output of C++: int x=11; cout << (x>10 ? "big" : "small"); ?

A. small

Q178. Output of C++: for(int i=0;i<4;i++) cout<<i; ?

A. 4

Q179. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q180. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 5

Q181. Which cast is safest for polymorphism in C++?

A. dynami

Q182. Template specialization allows?

A. No gen

Q183. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. big

Q184. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 543210

Q185. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ ha

Q186. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q187. Which cast is safest for polymorphism in C++?

A. reinterp

Q188. Template specialization allows?

A. Faster

Q189. Output of C++: int x=2; cout << (x>10 ? "big" : "small"); ?

A. error

Q190. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210

Q191. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch a

Q192. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Referer

Q193. Which cast is safest for polymorphism in C++?

A. const_c

Q194. Template specialization allows?

A. Only m

Q195. Output of C++: int x=15; cout << (x>10 ? "big" : "small"); ?

A. big

Q196. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120

Q197. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try alwa

Q198. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0

Q199. Which cast is safest for polymorphism in C++?

A. const_c

Q200. Template specialization allows?

A. No gen

ANSWER KEY

1-D, 2-D, 3-C, 4-B, 5-B, 6-D, 7-A, 8-D, 9-B, 10-C

11-C, 12-C